Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: *BLURB* | ***Context:*** *EU/French Pharmaceutical Regulatory NLP — BLURB Assessment*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

BLURB's output forms — categorical labels for classification, BIO/BIOUL sequence tags for NER, regression scores for STS, accuracy for QA, F1 variants for classification/extraction, Pearson correlation for STS [Q76, Q83, Q100, Q176] — broadly match the representational forms the deployment requires (entity tags, similarity/equivalence scores, classification labels). At this representational level the form mismatch is modest, and the deployment elicitation rates OF as MODERATE priority. However, BLURB reports only aggregate metrics: macro-averaged BLURB score [Q52, Q150], task-level F1/accuracy/Pearson, and averages across multiple runs for variance management [Q140]. There are no confidence-calibration metrics, threshold-sensitivity analyses, or human-review-triggering decision boundaries documented [Q150, Q151, Q152] — precisely the output-form information the deployment's human-review flagging workflow requires. The deployment's binary 'flag for human review' decision derives from continuous model outputs, but BLURB does not characterize model behavior at decision boundaries or report calibration. The form match is therefore partial: representational forms align, but evaluation forms (calibration, threshold sensitivity) do not.

ID	Evidence
Q52	“To compute a summary score for BLURB, the simplest way is to report the average score among all tasks. However, this may place undue emphasis on simpler tasks such as NER for which there are many existing datasets. Therefore, we group the datasets by their task types, compute the average score for each task type, and report the macro average among the task types.” (p.7)
Q76	“We use the same train/dev/test split in Peng et al. [45] and use Pearson correlation for evaluation.” (p.9)
Q83	“We instead adopt the original dataset and report micro F1 across the ten cancer hallmarks.” (p.9)
Q100	“We use cross-entropy loss for classification tasks and mean-square error for regression tasks.” (p.10)
Q140	“We report the average scores from ten runs for BIOSSES, BioASQ, and PubMedQA, and five runs for the others.” (p.13)
Q150	“The BLURB score is the macro average of average test results for each of the six tasks (NER, PICO, relation extraction, sentence similarity, document classification, question answering).” (p.14)
Q151	“We compare BERT models by applying them to the downstream NLP applications in BLURB.” (p.14)
Q152	“For each task, we conduct the same fine-tuning process using the standard task-specific model as specified in subsection 2.4.” (p.14)
Q176	“Comparison of entity-level F1 for biomedical named entity recognition (NER) using different tagging schemes and the standard PubMedBERT.” (p.19)

Strengths

- Output representational forms (categorical labels, sequence tags, regression scores) [Q100, Q107, Q176] directly map to the deployment's NER and STS output requirements
- BLURB provides standardized task-specific metrics (entity-level F1, micro F1, Pearson) [Q76, Q83, Q176] that are widely used and interpretable, easing methodological reuse
- Public release of pretrained and task-specific models alongside the leaderboard [Q3, Q196] supports reproducibility, which is valuable for any regulatory NLP system that must be auditable

ID	Evidence

Q3	“we have released our state-of-the-art pretrained and task-specific models for the community, and created a leaderboard featuring our BLURB benchmark (short for Biomedical Language Understanding & Reasoning Benchmark).” (p.1)
Q76	“We use the same train/dev/test split in Peng et al. [45] and use Pearson correlation for evaluation.” (p.9)
Q83	“We instead adopt the original dataset and report micro F1 across the ten cancer hallmarks.” (p.9)
Q100	“We use cross-entropy loss for classification tasks and mean-square error for regression tasks.” (p.10)
Q107	“BIO is the standard tagging scheme that classifies each token as the beginning of an entity (B), inside an entity (I), or outside (O).” (p.11)
Q176	“Comparison of entity-level F1 for biomedical named entity recognition (NER) using different tagging schemes and the standard PubMedBERT.” (p.19)
Q196	“To accelerate research in biomedical NLP, we release our state-of-the-art biomedical BERT models and setup a leaderboard based on BLURB.” (p.21)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (text labels, scores) matches the deployment's enterprise document-management integration; no speech, no mobile-output requirement is in scope. Representational match is good.

OF-2: Not applicable — the deployment is a desktop/server enterprise context with no speech-output requirement.

OF-3: Not applicable — end users are university-educated regulatory professionals; literacy and accessibility are not constraints.

OF-4: External validity is harmed at the evaluation-form level: BLURB does not report confidence calibration or threshold-sensitivity analyses [Q150, Q151], and the human-review flagging workflow requires precisely this information. Aggregate F1/Pearson scores do not characterize model behavior at the decision boundary the deployment uses.

ID	Evidence
Q100	“We use cross-entropy loss for classification tasks and mean-square error for regression tasks.” (p.10)
Q107	“BIO is the standard tagging scheme that classifies each token as the beginning of an entity (B), inside an entity (I), or outside (O).” (p.11)
Q150	“The BLURB score is the macro average of average test results for each of the six tasks (NER, PICO, relation extraction, sentence similarity, document classification, question answering).” (p.14)
Q151	“We compare BERT models by applying them to the downstream NLP applications in BLURB.” (p.14)
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WEB-6	EMA AI Reflection Paper requires human-oversight mechanisms and quality review for AI in product information drafting — implies threshold/calibration characterization the deployment must satisfy (https://www.ema.europa.eu/en/documents/scientific-guideline/reflection-paper-use-artificial-intelligence-ai-medicinal-product-lifecycle_en.pdf)

Information Gaps

- Whether the deployment's human-review flagging is implemented as a fixed score threshold or a calibrated probability cutoff

Requires Expert Verification

- Calibration and threshold-sensitivity reporting requirements implied by EU AI Act high-risk classification for the deployment

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Remediation

Gap 1: BLURB reports only aggregate metrics (macro-averaged scores, F1, Pearson) [Q150, Q176] and no calibration or threshold-sensitivity analysis, but the deployment requires confidence-based human-review triggering.

Recommendation: Augment evaluation with calibration metrics (ECE, reliability diagrams), threshold-sensitivity sweeps, and decision-boundary characterization against a regulatory-expert gold standard. Document these in line with EMA AI Reflection Paper expectations [WEB-6] and EU AI Act high-risk-system requirements.

ID	Evidence
Q150	"The BLURB score is the macro average of average test results for each of the six tasks (NER, PICO, relation extraction, sentence similarity, document classification, question answering)." (p.14)
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